

IDS 707: Data Visualization and Storytelling

Machine Learning Presentations

Overview

In your careers, you will often present your work to stakeholders who are not data scientists. These may be executives, program directors, policy makers, or professionals in other fields. Being able to communicate your technical work clearly, ethically, and persuasively to a targeted audience is an essential skill for data scientists.

For this final project, you will **reframe and present the results of your Principles of Machine Learning (IDS 705) final project**. You will adapt your presentation to a **non-technical stakeholder** with a strong interest in your project topic. The goal is to **communicate your work clearly, persuasively, and responsibly** to someone who cares about the outcomes but does not have machine learning expertise.

Each group will be assigned a **stakeholder persona**. This is a realistic professional who might work in a relevant department, be a senior decision-maker, or have domain expertise tied to your project. This persona represents your intended audience.

Important note: You are not redoing your machine learning analysis. Focus on **translating your technical findings into accessible insights**, emphasizing **why they matter, what they show, and how they can inform action or decisions**.

Learning Objectives

By completing this assignment, you will:

- Practice translating technical data science work into accessible insights for non-expert audiences.
- Develop your skills in **visual storytelling and clarity of communication**.
- Learn to **anticipate stakeholder needs, concerns, and follow-up questions**.
- Collaborate to **design and deliver a concise, focused presentation** that demonstrates professionalism and ethical awareness.

Task Summary

- Generate a **stakeholder persona** using the provided ChatGPT prompt.
- Prepare a **6-minute presentation** with supporting slides (PowerPoint, Google Slides, etc.).

- Tailor your **purpose, narrative, visualizations, and emphasis** to your assigned persona.
 - Your presentation should have a clearly stated **purpose**:
 - **Feedback:** request feedback on an approach or decision you have already made
 - **Decision-making:** The purpose is to have the stakeholder make a decision based on the information you are sharing
 - **Persuasion:** Persuade your stakeholder to take a particular action or particular point of view
 - Include a short set of **supplemental slides** that anticipate likely stakeholder questions or concerns.
 - Submit your slides **before the presentation date**.
 - You will present during the final class sessions.
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Why Revisit Your Machine Learning Project?

In real-world data science work, you are often asked to revisit past analyses and explain them to new audiences — a new manager, different team, or external stakeholder. You might also need to adapt your message because priorities, data, or decision contexts have changed.

This project gives you structured practice doing exactly that: taking an existing technical analysis and reframing it into a **clear, audience-centered, decision-relevant story**.

Deliverables and Deadlines

Date	Deliverable	Description
Fri, Nov 10*	Proposal	Submit project description and stakeholder persona for approval.
Sun, Nov 24	Slide Deck Due	Submit slides (link or file upload).
Mon, Nov 25	In-Class Presentations	Present during class (10:00 a.m.–12:45 p.m.).
Tues Nov 26	Individual reflections	Individual reflections submitted via Canvas

*You can submit your proposal anytime before Nov 10 for earlier approval, if you like.

Generate your persona

Each group will generate a **stakeholder persona** using the ChatGPT prompt below.

To do this:

1. Open a **new ChatGPT session**.
2. Paste the following prompt and insert your ML project description in the designated section.
3. Choose **one** of the two personas ChatGPT generates.
4. Submit the chosen persona with your proposal.

Please resist the temptation to rerun or edit the prompt multiple times to get a specific outcome. The point is to adapt your communication to *whichever plausible audience* you receive.

Prompt:

“I am going to share a description of a machine learning project I completed. Create two credible and specific personas who might have a stake in the topic. They can have domain knowledge or other relevant expertise, and have some basic understanding of data and analysis, but they should not have any knowledge of machine learning techniques or advanced statistics. The personas should work within an organization that would benefit from this project and be in a role that would reasonably expect to hear a presentation about this work.

For each persona, include a brief bulleted summary of the following:

- Role/Job title and organization
- Background (domain knowledge, technical knowledge, constraints)
- Goals
- Questions or concerns they might bring to a presentation on my topic
- Do not include demographic information.

Here is a summary of my machine learning project, including the topic, question, and what I did:

[Insert your project description here]”

Project Proposal

Submit the following as a single short document under the “ML Presentations Proposal” Canvas Assignment:

1. A brief **description of your original Machine Learning project** (100–200 words).
2. The **description of the stakeholder persona** you will use.
3. A **brief framing statement** (100–200 words) describing your presentation’s purpose and approach, explaining how your persona’s goals will shape your communication choices.

Proposals are not graded for points but will receive feedback on **scope, framing, and appropriateness of the persona**.

Slide Deck Requirements

Prepare a **6-minute presentation** with your group.

Slide Content & Structure

- The **first slide (before your title)** must include a **one-slide summary of your persona** (role, organization, constraints, etc.).
- Tailor your **purpose, story arc, and visuals** to this persona’s perspective.
- State the **purpose/goal** of your presentation (feedback, decision, or persuasion).
- Use an **hourglass or inverted pyramid storytelling structure**.
- Include at least **one data visualization** that clearly and ethically communicates key results.
- Apply best practices in **clarity, accessibility, and design** (labels, annotation, visual hierarchy).

Supplemental Slides (5–10 slides max):

- Anticipate **likely stakeholder questions** or follow-up topics.
 - Include additional visuals, details, or background that didn’t fit in the main presentation.
 - May contain **slightly more technical detail** if necessary for context.
 - Keep focused — not everything belongs in backup slides. Think strategically about the *most probable* questions or needs.
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Submission Instructions

Submit your slides by **link or upload** before your presentation:

- **Preferred:** Add a viewable Google Slides link to this shared spreadsheet: [Google Submission Sheet](#)
 - **Alternative:** Upload a PowerPoint file to Canvas.
 - Ensure all group members' names are on the title slide.
 - **Naming convention:** TeamName_IDS707_FinalProject.pptx
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Presentation Day

- **Date:** Monday, November 25, 10:00 a.m.–12:45 p.m.
- **Location:** Our regular classroom (reserved for the full session).
- Each team has **6 minutes to present** and **4 minutes for questions**.
- The first few seconds should introduce your **intended stakeholder persona**.
- Speak **as if your persona is in the room** — we'll act as members of their organization.
- Time penalties: **–2 points (out of 100)** for every 20 seconds over 6 minutes.
- You are expected to attend the full session and remain actively engaged for all presentations.

Participation

As with all class periods, this session counts toward your **overall participation grade**.

- You will be assessed on your **attention, engagement, and the quality of questions or comments** you provide to peers.
- There may not be time for everyone to ask verbal questions; you'll also submit a **written question or comment** after each presentation.

Individual Reflections

To accompany your group project, each student will submit an **individual reflection** (150–250 words) within 48 hours of the final presentation session.

In your reflection, address the following:

1. Your Role and Contribution

- Briefly describe your main contributions to the project and presentation.

2. Audience Adaptation

- How did the stakeholder persona influence your team's choices in framing, visuals, or language?
- Give one concrete example of how your approach changed because of your audience.

3. Personal Learning

- What did you learn about communicating technical or machine learning concepts to non-experts?
- Is there anything you would do differently next time you adapt technical work for a stakeholder audience?

The reflection will be submitted on Canvas under “ML Presentation Reflections”

Evaluation and Grading (100 points total)

Category	What We're Looking For	Points
Audience Fit & Framing	Presentation is clearly tailored to the assigned stakeholder persona; message aligns with their goals, constraints, and domain context. Technical ideas are communicated in plain, accurate language a non-specialist can follow	15
Clarity & Accessibility	Technical ideas are communicated in plain, accurate language a non-specialist can follow.	10

Category	What We're Looking For	Points
Storytelling & Structure	Presentation follows a clear, engaging structure (hourglass or inverted pyramid). Key takeaways are prioritized and well sequenced.	15
Slide Design	Slides are visually clean, readable, and support spoken content. They follow best practices of slide design discussed in class.	10
Data Visualizations	At least one visualization is included. Visualizations are used effectively, following principles discussed in class (clarity, accuracy, appropriate encoding, data ink, etc).	15
Team Delivery	All members contribute equally; pacing, transitions, and timing are smooth.	10
Documentation & Supplemental Slides	Follow-up slides thoughtfully anticipate stakeholder questions; content shows preparation for deeper discussion.	10
Ethical & Responsible Framing	Presentation acknowledges uncertainty, limitations, or responsible use considerations in a way that maintains credibility and trust.	10
Individual Reflections	Clearly describes individual role and contributions with specific examples, demonstrating meaningful intellectual contribution. Articulates how the persona influenced the team's communication decisions, using at least one concrete example.	5
